

Autoscaling in Kubernetes

Since you're learning **KCNA as a beginner**, let's understand Autoscaling in a very simple way.

1. What is Autoscaling?

Autoscaling means automatically increasing or decreasing resources according to the workload.

In Kubernetes, this usually means:

Low traffic



Fewer Pods

High traffic



More Pods

 **Easy memory:**

Autoscaling = Automatically adjust capacity according to demand.

2. Why do we need Autoscaling?

Imagine your website normally has:

2 Pods

But suddenly many users visit:



Two Pods may become overloaded.

Kubernetes can automatically increase the number:

Before:

Pod 1

Pod 2

After:

Pod 1

Pod 2

Pod 3

Pod 4

Pod 5

When traffic decreases:

5 Pods



2 Pods

This helps avoid wasting resources.

3. Kubernetes Autoscaling Types ★★ ★

For KCNA, remember **three important types**:

Autoscaling

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├— HPA

├— VPA

└— Cluster Autoscaler

4. HPA — Horizontal Pod Autoscaler ★

What does Horizontal mean?

Horizontal = Add or remove Pods.

2 Pods



High load



5 Pods

HPA automatically adjusts the **number of Pod replicas** based on metrics.

For example:

CPU usage > target



More Pods

When demand decreases:

CPU usage ↓



Fewer Pods

 **Memory:**

HPA = More/Fewer Pods

5. VPA — Vertical Pod Autoscaler ★

What does Vertical mean?

Vertical = Give an existing Pod more or less resources.

For example:

Pod

CPU: 500m

Memory: 512Mi

If the workload needs more resources, VPA can recommend or adjust resource requests/limits depending on its configuration.

Conceptually:

Small resources



More resources



Pod

🧠 **Memory:**

VPA = More/Less resources per Pod

6. Cluster Autoscaler ★

HPA and VPA deal with **Pods/resources for workloads**.

But what if the cluster itself doesn't have enough Nodes?

That's where **Cluster Autoscaler** comes in.

Example:

Cluster

|

|— Node 1

└— Node 2

Pods need more capacity:

Not enough space ❌



Cluster Autoscaler



Add Node 3

When Nodes are no longer needed:

Node 3



No longer needed



Can be removed

 **Memory:**

Cluster Autoscaler = Adds/removes Nodes

7. HPA vs VPA vs Cluster Autoscaler

Autoscaler	Changes	Easy Meaning
HPA	Number of Pods	⊕/⊖ Pods
VPA	Resources of Pods	⬆/⬇ CPU & Memory
Cluster Autoscaler	Number of Nodes	⊕/⊖ Nodes

★ Super Easy Trick

HPA → Pods

VPA → Resources

CA → Nodes

8. Real-Life Example

Imagine a supermarket.

HPA

More customers arrive:

1 cashier



4 cashiers

 **More Pods**

VPA

One cashier needs a bigger workspace:

Small counter



Large counter

 **More resources for a Pod**

Cluster Autoscaler

The supermarket itself needs more space:

1 building



2 buildings

 **More Nodes**

9. Important Difference: Scaling Up vs Scaling Out

Scaling Out

Add more instances/Pods:

Pod → Pod → Pod → Pod

This is **horizontal scaling**.

Scaling Up

Give an existing Pod more CPU/memory:

Pod



More CPU/RAM

This is **vertical scaling**.

 **Remember:**

Horizontal = More Pods

Vertical = Bigger/More resources

★ KCNA Revision Notes

Autoscaling

- **Autoscaling = automatically adjusting capacity based on demand.**
- **HPA** → increases/decreases the number of Pods.
- **VPA** → adjusts/recommends CPU and memory resources for Pods.
- **Cluster Autoscaler** → increases/decreases the number of Nodes.
- Horizontal scaling = **more Pods**.
- Vertical scaling = **more resources per Pod**.

 **One-line memory:**

HPA → How many Pods? 

VPA → How much CPU/RAM? 

CA → How many Nodes? 

The most important KCNA point:

HPA scales Pods horizontally, VPA deals with Pod resources vertically, and Cluster Autoscaler scales the cluster's Nodes.